

## **Bridging Neuroscience and Artificial Intelligence**

Student Name: Win Aung

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Instructor's Name: Professor Patricia Mcmanus

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## **Table of Contents**

Part 1: Neural Network Comparison Analysis

Part 2: Brain Architecture and AI Design

Part 3: Integration Analysis and Future Implications

References

Part 1: Neural Network Comparison Analysis

Biological Neuron Structure

Using Wolfram Alpha's neuron anatomy resources, the following structural components of a biological neuron were identified and analyzed for their functional roles.

**Cell Body (Soma):** The soma is the metabolic center of the neuron. It integrates incoming signals from thousands of synaptic connections and generates an action potential when the combined input exceeds a threshold of approximately -55 mV. In artificial neural networks, this maps to the processing unit, which computes a weighted sum of inputs and applies an activation function.

**Dendrites:** Dendrites are branched extensions that receive electrochemical signals from other neurons. A single neuron may have thousands of dendritic branches with specialized receptor proteins. In artificial networks, dendrites correspond to input connections or weighted edges carrying data from previous layers.

**Axon:** The axon is a long projection that transmits the action potential away from the cell body toward target neurons. Myelination enables rapid conduction at speeds up to 120 m/s. In artificial networks, the axon maps to the output connection carrying the neuron's activation to downstream units.

**Synapses:** Synapses are the junctions where signal transmission occurs between neurons. The efficacy of these connections changes through long-term potentiation and long-term depression, which is the biological basis of learning. This corresponds to weights in artificial networks, adjusted during training through backpropagation.

Component Mapping

| Biological Component | Function                                   | Artificial Counterpart | AI Function                                |
|----------------------|--------------------------------------------|------------------------|--------------------------------------------|
| Cell body (soma)     | Integrates signals; fires action potential | Processing unit        | Weighted sum + activation function         |
| Dendrites            | Receive input signals                      | Input connections      | Weighted edges from previous layer         |
| Axon                 | Transmits output signal                    | Output connection      | Passes activation to next layer            |
| Synapses             | Modifiable junctions; basis of learning    | Weights                | Learnable parameters adjusted via training |
| Neurotransmitters    | Chemical signal modulation                 | Activation functions   | ReLU, sigmoid, tanh                        |

**Comparison: Biological vs. Artificial Neurons**

The design of artificial neurons was directly inspired by biological neural processing. Rosenblatt's perceptron (1958) modeled the neuron's integrative function as a weighted sum passed through a threshold, mirroring the soma's role in generating action potentials. However, the two systems differ substantially. Biological neurons employ asynchronous, spike-based coding where information is carried in the timing and frequency of electrical pulses, whereas artificial neurons operate synchronously with continuous-valued activations through layer-by-layer computation.

The human brain operates on roughly 20 watts yet performs massively parallel computation across 86 billion neurons, each averaging 7,000 synaptic connections. Training large artificial networks consumes kilowatts to megawatts of power. Biological systems leverage multiple learning mechanisms including Hebbian plasticity, synaptic tagging, and neuromodulation, enabling continuous unsupervised adaptation. Artificial systems rely on distinct training phases using gradient-based optimization.

Each approach offers advantages: biological neurons excel at energy efficiency, generalization from few examples, and graceful degradation under damage. Artificial neurons provide mathematical precision, exact reproducibility, and scalable deployment across digital hardware.

**Comparative Analysis Table**

| Feature         | Biological Neural Network        | Artificial Neural Network          | AI Example                     |
|-----------------|----------------------------------|------------------------------------|--------------------------------|
| Signal type     | Electrochemical spikes           | Continuous floating-point values   | Spiking Neural Networks        |
| Processing      | Massively parallel, asynchronous | Layer-by-layer, synchronous        | GPU-parallelized DNNs          |
| Learning rule   | Hebbian plasticity (LTP/LTD)     | Backpropagation / gradient descent | Adam optimizer in Transformers |
| Energy          | ~20 W for entire brain           | Kilowatts to megawatts             | GPT-3 training: ~1,300 MWh     |
| Connectivity    | ~7,000 synapses/neuron, sparse   | Fully connected or structured      | CNN local receptive fields     |
| Fault tolerance | High, graceful degradation       | Variable, dropout helps            | Dropout regularization         |
| Memory          | Distributed across synapses      | Stored in weight matrices          | LSTM gated memory cells        |

|                |                              |                                |                            |
|----------------|------------------------------|--------------------------------|----------------------------|
| Adaptability   | Continuous lifelong learning | Typically fixed after training | Continual learning / EWC   |
| Attention      | Selective neural gating      | Attention weight computation   | Transformer self-attention |
| Generalization | Excellent from few examples  | Requires large datasets        | Few-shot learning in GPT-4 |

## Part 2: Brain Architecture and AI Design

### Hierarchical Brain Organization

Using Wolfram Alpha's brain anatomy resources, the human brain reveals a hierarchical processing architecture that has directly influenced deep learning design. The cortex is organized into distinct levels that progressively transform raw sensory input into abstract representations.

**Primary sensory areas (V1)** detect basic features such as edges and orientations. **Secondary visual areas (V2, V4)** integrate these into shapes and color patterns. The **inferotemporal cortex** performs high-level object recognition. The **prefrontal cortex** handles abstract reasoning and decision-making.

The brain also shows regional specialization: the occipital lobe handles visual processing, the temporal lobe manages auditory processing and memory, the parietal lobe controls spatial awareness, the frontal lobe governs motor planning and decisions, and the hippocampus consolidates memory.

### Brain-AI Architecture Parallels

**CNNs and the Visual Cortex:** Hubel and Wiesel's (1962) work on simple and complex cells in the visual cortex directly inspired convolutional neural networks. Simple cells respond to edges at specific orientations within small receptive fields, paralleling convolutional filters in early CNN layers. Complex cells exhibit spatial invariance, corresponding to pooling operations. The hierarchical progression from V1 through V4 to IT mirrors the layer-by-layer feature abstraction in architectures like ResNet.

**LSTMs and Hippocampal Memory:** The hippocampus selectively consolidates important information into long-term memory while allowing irrelevant details to fade. This parallels the gating architecture of LSTM networks. The forget gate determines which information to discard, analogous to hippocampal memory filtering. The input gate controls what new information to store, similar to hippocampal encoding. Both solve the challenge of maintaining relevant information across long temporal sequences.

**Attention Mechanisms:** The brain's attention system, governed by the frontal eye fields and posterior parietal cortex, selectively enhances responses to relevant stimuli. The Transformer's self-attention mechanism (Vaswani et al., 2017) mirrors this by computing relevance scores between all input elements, enabling dynamic allocation of processing resources regardless of position.

### Amygdala and Emotion Processing

The amygdala is central to emotion processing and threat detection. Its dual-pathway architecture, with a fast subcortical route for rapid responses and a slower cortical route for detailed evaluation, offers insights for AI design.

**Decision Making:** The dual-pathway processing parallels Kahneman's (2011) System 1 and System 2 framework. AI systems could deploy a lightweight rapid-response module for time-critical decisions alongside a thorough analytical module for complex situations.

**Risk Assessment:** The amygdala's threat detection suggests AI safety systems could benefit from dedicated anomaly detection modules that flag unusual patterns before full analysis, enhancing robustness in critical applications.

**Pattern Recognition:** The amygdala tags experiences with emotional valence, enhancing memory consolidation for significant events. AI systems could modulate learning rates based on prediction error magnitude, improving adaptability and efficiency.

## Part 3: Integration Analysis and Future Implications

### Improving AI Through Biological Insights

The human brain performs complex cognitive tasks on approximately 20 watts while state-of-the-art AI training consumes megawatts, demonstrating significant room for improvement. Neuromorphic computing, implementing spiking neural networks on specialized hardware like Intel's Loihi chip, represents one promising approach. By adopting the brain's event-driven processing, these systems achieve dramatic energy savings while maintaining competitive performance.

Biological insights also address persistent AI challenges. Catastrophic forgetting, where networks lose previous knowledge when learning new tasks, has a biological counterpart in synaptic consolidation, where important connections are selectively stabilized. Methods inspired by this, such as Elastic Weight Consolidation, show promise for continual learning.

Hawkins' Thousand Brains Theory (2021) proposes that the neocortex consists of thousands of cortical columns, each building complete models through reference frames. This columnar voting architecture suggests new approaches to robust AI design. Additionally, the brain's predictive coding framework, where each cortical level generates predictions and propagates only prediction errors, offers a computationally efficient alternative to purely feedforward processing in standard Transformer models.

### Critical Comparison

**Processing Efficiency:** Biological systems are vastly superior in energy efficiency with ~20 W for all cognitive functions versus kilowatts for AI training. The gap is narrowing with neuromorphic hardware.

**Learning Capabilities:** Biological learning is more flexible with few-shot learning and continuous adaptation. AI excels at precise optimization on defined objectives but typically requires large labeled datasets.

**Adaptability:** Biological systems support lifelong learning and can compensate for damage. AI systems are typically fixed after training, and continual learning remains a major challenge.

**Generalization:** Biological systems generalize excellently from limited experience. Foundation models like GPT-4 are narrowing this gap but still struggle with out-of-distribution scenarios.

### Future Directions

Understanding the brain's binding problem, how it integrates multiple sensory modalities into unified experiences, could inform multimodal AI systems. The brain's complementary memory systems, with fast episodic memory in the hippocampus and slow statistical learning in the



neocortex, suggest hybrid architectures combining rapid adaptation with gradual consolidation. Retrieval-augmented generation systems represent early steps in this direction.

Research into biologically plausible learning rules, combining local Hebbian updates with global modulatory signals, could yield training algorithms that are more energy-efficient and suited to neuromorphic hardware. Advances in brain-computer interfaces are also creating possibilities for hybrid systems leveraging the strengths of both biological and artificial intelligence.

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